

human

Which houses in Exeter are 3m or less above sea level?

host_agent

Tool call: send_message - target: planning_agent - task_description: User wants to find houses in Exeter that are 3m or less above sea level. Provide the general steps to solve this geospatial query.

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are houses in Exeter, and the condition is that these houses must be 3m or less above sea level.

output steps: ["Find Exeter 1 area", "Find houses in Exeter as many search results", "Apply condition for houses that are 3m or less above sea level"]

host_agent

Tool call: send_message - target: named_area - task_description: Find the area polygon for Exeter.

named_area_agent

Tool call: call_os_ngd - filters: None - bbox: None - polygon_or_point: True - street_address: None - filename: exeter_area_polygon

Tool call result: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.., count of records fetched : 98.

Tool call: send_message - target: data_analysis_agent - task_description: Filter the artifact 'exeter_area_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon.'

data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['exeter_area_polygon']

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where the area is specifically for Exeter
    # Use multiple columns for accuracy: name1_text, description, descriptiongroup
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter', na=False))) &
                  ((gdf['description'].str.lower().str.contains('city', na=False)) |
                   (gdf['descriptiongroup'].str.lower().str.contains('settlement', na=False)))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter area."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing the Exeter area, filtered using name and description columns.'
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered to 1 polygon for the Exeter area. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) representing the Exeter area, filtered using both the name and description columns for accuracy.

named_area_agent

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) representing the Exeter area.

host_agent

Tool call: send_message - target: buildings - task_description: Find all houses (residential buildings) within the 'filtered_exeter_area_polygon' artifact.

buildings_agent

Tool call: call_os_ngd - filters: None - bbox: filtered_exeter_area_polygon - polygon_or_point: True - street_address: False - filename: exeter_residential_buildings

Tool call result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_exeter_residential_buildings', 'buildingline_exeter_residential_buildings', 'buildingpart_exeter_residential_buildings'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find all houses (residential buildings) within the 'filtered_exeter_area_polygon' artifact. using filters None.", ...], counts of records fetched are : [69169, 69342, 85480].

Tool call: send_message - target: data_analysis_agent - task_description: Filter the artifact 'buildings_exeter_residential_buildings' to find houses defined as buildings where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Save the filtered results as 'filtered_houses_exeter'. Provide a summary of the findings.

data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['buildings_exeter_residential_buildings']

Tool Code Executor

```
def filter_houses_exeter(data: list):
    gdf = data[0]
    # Filter for houses: residential address count > 0 and total address count == 1
    filtered = gdf[(gdf['buildinguse_addresscount_residential'] > 0) & (gdf['buildinguse_addresscount_total'] == 1)]
    summary = f"Filtered houses in Exeter: {len(filtered)} entries found out of {len(gdf)} total buildings."
    artifact_name = 'filtered_houses_exeter'
    artifact_description = (
        "A GeoDataFrame containing buildings in Exeter where the number of residential addresses is greater than 0 "
        "and the total address count is exactly 1. This typically identifies single-household residential buildings."
    )
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered houses in Exeter: 38,848 entries were found that meet the criteria (buildings with more than 0 residential addresses and a total address count of exactly 1), out of a total of 69,169 buildings.

A data artifact named 'filtered_houses_exeter' has been generated, containing these single-household residential buildings in Exeter. If you need further analysis or details on this filtered set, please let me know!

buildings_agent

I found 38,848 houses in the 'filtered_exeter_area_polygon' that meet the criteria for single-household residential buildings. The filtered results are saved in the artifact named 'filtered_houses_exeter'. If you need further analysis or details, feel free to ask!

host_agent

Tool call: send_message - target: plotting_agent - task_description: From the 'filtered_houses_exeter' artifact, identify and plot the houses that are 3m or less above sea level within the 'filtered_exeter_area_polygon'. Apply the elevation condition and generate a summary.

plotting_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['filtered_houses_exeter', 'filtered_exeter_area_polygon']

Tool Code Executor

```
def plot_low_elevation_houses(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    houses = data[0].copy()
    exeter_poly = data[1].copy()

    # Ensure CRS is EPSG:4326 for folium
    if houses.crs is not None and houses.crs.to_epsg() != 4326:
        houses = houses.to_crs(epsg=4326)
    if exeter_poly.crs is not None and exeter_poly.crs.to_epsg() != 4326:
        exeter_poly = exeter_poly.to_crs(epsg=4326)

    # Filter houses within the Exeter polygon
    houses_in_exeter = gpd.sjoin(houses, exeter_poly, predicate='within', how='inner')

    # Filter for houses with minimum elevation (height_absolutemin_m) <= 3m
    low_elev_houses = houses_in_exeter[houses_in_exeter['height_absolutemin_m'] <= 3]

    # Prepare folium map centered on Exeter
    exeter_centroid = exeter_poly.geometry.iloc[0].centroid
    m = folium.Map(location=[exeter_centroid.y, exeter_centroid.x], zoom_start=13)

    # Add Exeter polygon
    folium.GeoJson(mapping(exeter_poly.geometry.iloc[0]),
                    name='Exeter Area',
                    style_function=lambda x: {'fillColor': '#00000000', 'color': 'blue', 'weight': 2}).add_to(m)

    # Add low elevation houses
    for _, row in low_elev_houses.iterrows():
        folium.GeoJson(row['geometry'],
                        style_function=lambda x: {'fillColor': '#ff0000', 'color': '#ff0000', 'weight': 1, 'fillOpacity': 0.5},
                        tooltip=f"ID: {row['osid']}<br>Min Elevation: {row['height_absolutemin_m']}m").add_to(m)

    # Save map
    map_filename = 'low_elevation_houses_exeter.html'
    m.save(map_filename)

    # Prepare summary
```

```

n = len(low_elev_houses)
summary_rows = low_elev_houses[['osid', 'height_absolutemin_m', 'description']].head(5).to_dict('records')
summary = f"Found {n} houses in Exeter with minimum elevation <= 3m above sea level. First 5: {summary_rows}"

return [summary, map_filename, "Folium map showing houses <= 3m above sea level within Exeter area polygon.", map_filename]

```

Error: KeyError: 'osid'

Tool Code Executor

```

def plot_low_elevation_houses(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    houses = data[0].copy()
    exeter_poly = data[1].copy()

    # Ensure CRS is EPSG:4326 for folium
    if houses.crs is not None and houses.crs.to_epsg() != 4326:
        houses = houses.to_crs(epsg=4326)
    if exeter_poly.crs is not None and exeter_poly.crs.to_epsg() != 4326:
        exeter_poly = exeter_poly.to_crs(epsg=4326)

    # Filter houses within the Exeter polygon
    houses_in_exeter = gpd.sjoin(houses, exeter_poly, predicate='within', how='inner')

    # Filter for houses with minimum elevation (height_absolutemin_m) <= 3m
    low_elev_houses = houses_in_exeter[houses_in_exeter['height_absolutemin_m'] <= 3]

    # Prepare folium map centered on Exeter
    exeter_centroid = exeter_poly.geometry.iloc[0].centroid
    m = folium.Map(location=[exeter_centroid.y, exeter_centroid.x], zoom_start=13)

    # Add Exeter polygon
    folium.GeoJson(mapping(exeter_poly.geometry.iloc[0]),
                    name='Exeter Area',
                    style_function=lambda x: {'fillColor': '#00000000', 'color': 'blue', 'weight': 2}).add_to(m)

    # Add low elevation houses
    for _, row in low_elev_houses.iterrows():
        folium.GeoJson(row['geometry'],
                        style_function=lambda x: {'fillColor': '#ff0000', 'color': '#ff0000', 'weight': 1, 'fillOpacity': 0.5},
                        tooltip=f"Min Elevation: {row['height_absolutemin_m']}m").add_to(m)

    # Save map

```

```
map_filename = 'low_elevation_houses_exeter.html'
m.save(map_filename)

# Prepare summary
n = len(low_elev_houses)
# Use available columns for summary
summary_cols = [col for col in ['height_absolutemin_m', 'description'] if col in low_elev_houses.columns]
summary_rows = low_elev_houses[summary_cols].head(5).to_dict('records')
summary = f"Found {n} houses in Exeter with minimum elevation <= 3m above sea level. First 5: {summary_rows}"

return [summary, map_filename, "Folium map showing houses <= 3m above sea level within Exeter area polygon.", map_filename]
```

Found 5 houses in Exeter with minimum elevation <= 3m above sea level. First 5: [{'height_absolutemin_m': 2.4}, {'height_absolutemin_m': 2.0}, {'height_absolutemin_m': 2.6}, {'height_absolutemin_m': 2.8}, {'height_absolutemin_m': 3.0}]

host_agent

There are 5 houses in Exeter that are 3 meters or less above sea level. These houses have minimum elevations ranging from 2.0m to 3.0m above sea level.

A map has been generated showing the locations of these houses within the Exeter area. If you would like to view or download this map, let me know!